

# Introduction to Electrical Engineering Practice Exam

Discussion: January 12, 2026

---

## 1 Conceptual Understanding: Multiple Choice

Select exactly one answer per question.

### Fundamentals

1. The drift motion of electrons in a conductor is mainly caused by:

- A) temperature gradients
- B) magnetic fields
- C) the electric field
- D) mechanical forces

Solution: C

2. For a metallic conductor, an increase in temperature usually leads to:

- A) decreasing resistance
- B) sudden insulating behavior
- C) unchanged resistance
- D) increasing resistance

Solution: D

### DC

3. Kirchhoff's current law states:

- A) The sum of voltages in a loop is zero.
- B) The sum of currents at a node is zero.
- C) The sum of resistances at a node is constant.
- D) Power is stored at a node.

Solution: B

4. An ideal voltage source is characterized by:

- A) a fixed voltage independent of current

- B) a fixed current independent of the load
- C) infinite internal resistance
- D) no influence on connected circuits

Solution: A

5. The nodal-voltage method primarily works with:

- A) mesh currents
- B) node voltages
- C) equivalent sources
- D) frequency analysis

Solution: B

6. For maximum power transfer from a source to a purely resistive load:

- A) load resistance  $\gg$  source resistance
- B) load resistance  $\ll$  source resistance
- C) load resistance equals source internal resistance
- D) load resistance equals zero

Solution: C

7. An ammeter is typically connected:

- A) in parallel with the measured quantity
- B) in series with the measured quantity
- C) with zero internal resistance
- D) only in AC circuits

Solution: B

## Fields & Components

8. A capacitor primarily stores:

- A) magnetic energy
- B) mechanical energy
- C) thermal energy
- D) electric energy

Solution: D

9. Under constant DC voltage, a capacitor behaves (in steady state) like:

- A) a short circuit
- B) an open circuit (no current flow)
- C) a resistor with fixed value
- D) a voltage source

Solution: B

**10.** Magnetic fields are mainly generated by:

- A) stationary charges
- B) moving charges or electric current
- C) temperature changes
- D) light radiation

Solution: B

**11.** Electrical induction occurs when:

- A) the voltage remains constant
- B) the conductor length remains constant
- C) the magnetic flux through an area changes
- D) no charges are present

Solution: C

**12.** An inductor opposes rapid changes in current because it:

- A) stores electric energy
- B) stores magnetic energy
- C) generates heat
- D) neutralizes charges

Solution: B

**13.** An ideal transformer can:

- A) convert DC voltage to higher DC voltage
- B) convert AC voltage to AC voltage with different amplitude
- C) permanently change the frequency of AC voltage
- D) generate energy

Solution: B

**14.** A transient process in an electrical network refers to:

- A) an instantaneous change without time dependence
- B) a time-dependent transition toward steady state
- C) a permanent state without variation
- D) a purely mathematical concept without physical meaning

Solution: B

## AC

15. A phasor represents:

- A) the complete time-domain waveform
- B) DC quantities only
- C) only the real part of a quantity
- D) a complex amplitude (magnitude and phase) at fixed frequency

Solution: D

16. The reactance of a capacitor decreases with increasing frequency:

- A) increases
- B) decreases
- C) remains constant
- D) is unpredictable

Solution: B

17. Reactive power occurs in an AC circuit when:

- A) voltage and current are phase-shifted
- B) voltage and current are exactly in phase
- C) no energy is transferred
- D) only DC current flows

Solution: A

18. A high-pass filter ideally allows:

- A) low frequencies
- B) no signals
- C) DC only
- D) high frequencies

Solution: D

**19.** Resonance in a resonant circuit typically means:

- A) minimum impedance / maximum current (series circuit)
- B) maximum damping
- C) no load influence
- D) instantaneous decay

Solution: A

**20.** A practical advantage of a symmetrical three-phase system is:

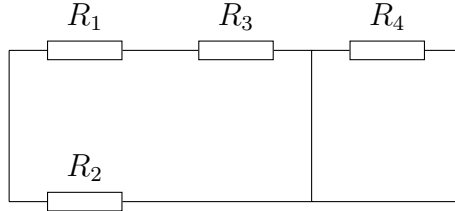
- A) lower frequency
- B) only two conductors are required
- C) generation of a rotating magnetic field (e.g. for motors)
- D) no balancing current possible

Solution: C

## 2 DC Networks

### 2.1 Equivalent Resistance

Given the resistor network with  $R_1 = 10\ \Omega$ ,  $R_2 = 20\ \Omega$ ,  $R_3 = 30\ \Omega$ ,  $R_4 = 15\ \Omega$ :



Calculate the equivalent resistance of the network.

Solution:

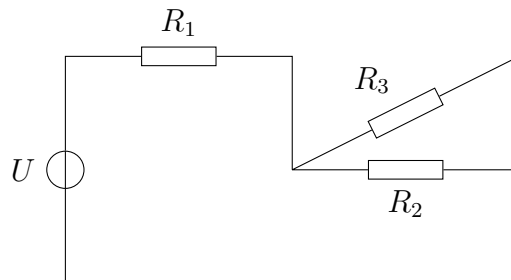
$$R_{12} = \frac{R_1 R_2}{R_1 + R_2} = \frac{10 \cdot 20}{30} = 6.67\ \Omega$$

$$R_{123} = 6.67 + 30 = 36.67\ \Omega$$

$$R = \frac{36.67 \cdot 15}{36.67 + 15} \approx 10.7\ \Omega$$

### 2.2 Network Analysis

Given the circuit with  $R_1 = 6\ \Omega$ ,  $R_2 = 12\ \Omega$ ,  $R_3 = 4\ \Omega$  and a voltage source of  $24\ \text{V}$ :



Determine:

1. the total source current,
2. the branch currents through  $R_2$  and  $R_3$ ,
3. the power dissipated in  $R_3$ .

Solution:

$$R_{23} = \frac{12 \cdot 4}{16} = 3\ \Omega$$

$$R = 6 + 3 = 9\ \Omega$$

$$I = \frac{24}{9} = 2.67\ \text{A}$$

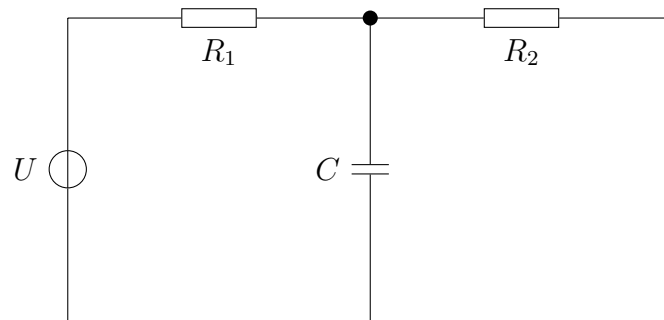
$$U_{23} = I \cdot 3 = 8\ \text{V}$$

$$I_2 = \frac{8}{12} = 0.67\ \text{A}, \quad I_3 = \frac{8}{4} = 2\ \text{A}$$

$$P_3 = I_3^2 R_3 = 16\ \text{W}$$

## 2.3 Nodal Analysis

Given the DC network with  $U = 20\text{ V}$ ,  $R_1 = 5\ \Omega$ ,  $R_2 = 10\ \Omega$ ,  $C = 100\ \mu\text{F}$ :



The lower conductor is chosen as reference potential (ground). The steady-state condition ( $t \rightarrow \infty$ ) is assumed.

Determine:

1. the node voltage at the marked node,
2. the voltage across the capacitor,
3. the current through the capacitor,
4. the energy stored in the capacitor.

Solution:

$$\frac{V_1 - 20}{R_1} + \frac{V_1 - 0}{R_2} = 0$$

$$2(V_1 - 20) + V_1 = 0$$

$$V_1 = \frac{40}{3} \approx 13.33\text{ V}$$

$$U_C = V_1 - 0 = 13.33\text{ V}$$

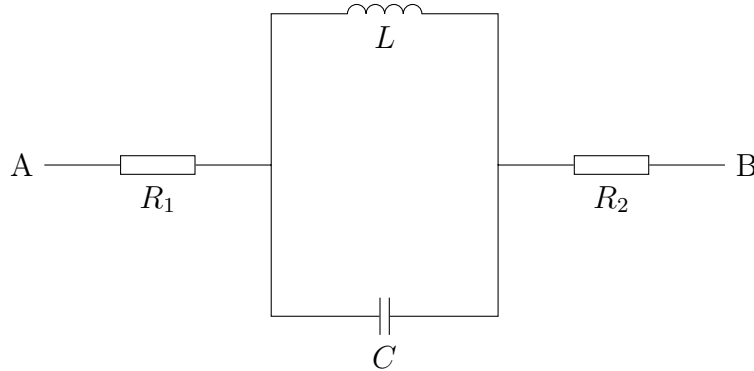
$$i_C = C \frac{du_C}{dt} = 0$$

$$W = \frac{1}{2} C U_C^2 \approx 8.9\text{ mJ}$$

## 3 AC Networks

### 3.1 Equivalent Impedance

Given the AC network with  $R_1 = 10\ \Omega$ ,  $R_2 = 20\ \Omega$ ,  $L = 50\text{ mH}$ ,  $C = 100\ \mu\text{F}$ ,  $\omega = 1000\text{ rad/s}$ :



Determine:

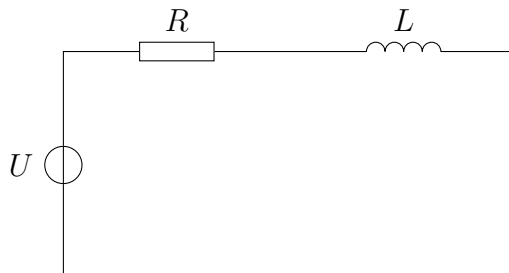
1. the complex impedances of the individual components,
2. the complex equivalent impedance between terminals A and B,
3. the magnitude and phase of the equivalent impedance.

Solution:

$$\begin{aligned}\underline{Z}_R &= R \\ \underline{Z}_L &= j\omega L = j \cdot 1000 \cdot 0.05 = j50\ \Omega \\ \underline{Z}_C &= \frac{1}{j\omega C} = \frac{1}{j \cdot 1000 \cdot 100 \cdot 10^{-6}} = -j10\ \Omega \\ \underline{Z}_{LC} &= \left( \frac{1}{\underline{Z}_L} + \frac{1}{\underline{Z}_C} \right)^{-1} = \frac{1}{j0.08} = -j12.5\ \Omega \\ \underline{Z} &= R_1 + \underline{Z}_{LC} + R_2 = (30 - j12.5)\ \Omega \\ |\underline{Z}| &= \sqrt{30^2 + 12.5^2} \approx 32.5\ \Omega \\ \varphi &= \arctan\left(\frac{-12.5}{30}\right) \approx -22.6^\circ\end{aligned}$$

### 3.2 Complex Network Analysis

Given the following network with a sinusoidal voltage source  $U = 230\text{ V}$ ,  $f = 50\text{ Hz}$  and the components  $R = 20\ \Omega$ ,  $L = 100\text{ mH}$ :





Determine:

1. the complex impedance  $Z$ ,
2. the RMS current  $I$ ,
3. the phase shift between current and voltage,
4. the power factor  $\cos \varphi$ ,
5. the active power  $P$ ,
6. the reactive power  $Q$ .

Solution:

$$\omega = 2\pi f = 314 \text{ rad/s}$$

$$X_L = \omega L = 31.4 \Omega$$

$$\underline{Z} = 20 + j31.4 \Rightarrow |Z| = 37.3 \Omega$$

$$I = \frac{230}{37.3} = 6.17 \text{ A}$$

$$\varphi = \arctan\left(\frac{31.4}{20}\right) = 57.5^\circ$$

$$\cos \varphi = 0.54$$

$$P = UI \cos \varphi = 766 \text{ W}$$

$$Q = UI \sin \varphi = 1180 \text{ var}$$

# Formula Sheet

## Capacitor

$$i_C(t) = C \frac{du_C(t)}{dt}$$

$$W = \frac{1}{2}CU^2$$

## AC

$$\omega = 2\pi f$$

$$\underline{Z}_L = j\omega L$$

$$\underline{Z}_C = \frac{1}{j\omega C}$$

$$\underline{Z} = R + jX$$

$$P = UI \cos \varphi$$

$$Q = UI \sin \varphi$$